Functional Annotation of *hisB* (Q88R45, PP_0289) in *Pseudomonas putida* KT2440

Gene: hisB | **Locus:** PP_0289 | **UniProt:** Q88R45 **Protein:** Imidazoleglycerol-phosphate dehydratase (IGPD), EC 4.2.1.19 **Organism:** *Pseudomonas putida* strain ATCC 47054 / DSM 6125 / KT2440 (PSEPK)

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1. Summary (Answer to the Research Question)

hisB (PP_0289) encodes **imidazoleglycerol-phosphate dehydratase (IGPD; EC 4.2.1.19)**, a cytoplasmic, manganese(II)-dependent hydro-lyase that catalyzes the **sixth step of the nine-step L-histidine biosynthetic pathway**. It removes water from D-*erythro*-1-(imidazol-4-yl)glycerol 3-phosphate (imidazoleglycerol phosphate, IGP) to yield 3-(imidazol-4-yl)-2-oxopropyl phosphate (imidazoleacetol phosphate, IAP). In *P. putida* the enzyme is the **monofunctional** form (197 aa, a single IGPD/PF00475 domain), in contrast to the bifunctional HisB of *Escherichia coli*; the downstream histidinol-phosphate phosphatase activity is supplied by a **separate gene, PP_3157**. The enzyme operates in the bacterial cytosol and, because the histidine pathway is absent in mammals, IGPD is a validated herbicide/antimicrobial target.

Gene-identity verification: CONFIRMED. UniProt Q88R45 (hisB/PP_0289, 197 aa, single IGPD domain, EC 4.2.1.19, cytoplasm) matches KEGG ppu:PP_0289 (K01693, IGPD) and the family/domain signatures (IPR000807, IPR020565, PF00475). All identifiers are internally consistent; there is no gene-symbol ambiguity for this protein.

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2. Primary Function: Reaction Catalyzed and Substrate Specificity

Reaction (EC 4.2.1.19):

> D-*erythro*-1-(imidazol-4-yl)glycerol 3-phosphate → 3-(imidazol-4-yl)-2-oxopropyl phosphate + H_2O

This is a **dehydration (hydro-lyase)** reaction: elimination of the C-2′ hydroxyl (as water) from the glycerol-phosphate side chain of the imidazole substrate. The product, imidazoleacetol phosphate, is subsequently transaminated by histidinol-phosphate aminotransferase (HisC/PP_0967) to histidinol phosphate.

Substrate specificity. IGPD is highly specific for its physiological substrate IGP; recognition is dominated by the imidazole ring, which is anchored between the two active-site Mn(II) ions, while the phosphate and hydroxyl groups are held by a network of conserved basic and acidic residues. In the *Mycobacterium tuberculosis* IGPD–IGP complex the substrate contacts Glu21, Arg99, Glu180, Arg121 and Lys184 contributed by three different protomers (Ahangar et al., 2013, P 24311587). The imidazole-focused recognition also explains inhibition by the imidazole/triazole mimic **3-amino-1,2,4-triazole (ATZ)**, which is a competitive inhibitor (P 24311587), and by triazole-phosphonate herbicides (Glynn et al., 2005, P 16338409).

Cofactor requirement. IGPD is a **Mn(II) metalloenzyme**; two manganese ions per active site are essential both for catalysis and for assembly of the functional oligomer (P 16338409).

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3. Catalytic Mechanism (Structural/Mechanistic Evidence)

Although no *P. putida* IGPD structure has been solved, the mechanism is well established from high-resolution structures of orthologs that share the same fold and conserved residues:

- **Quaternary structure:** The active enzyme is a **24-mer with 432 point-group symmetry**, containing **24 equivalent active sites**, each built at the interface of three protomers. Assembly of the 24-mer from an inactive trimeric precursor is triggered by formation of a **dimanganese cluster** (*Arabidopsis thaliana* IGPD; Glynn et al., 2005, P 16338409). The same 24-meric architecture was confirmed for *M. tuberculosis* IGPD (P 24311587).
- **Chemical mechanism:** The substrate's imidazole ring binds between the two Mn(II) ions as an **imidazolate**, which collapses to a high-energy **diazafulvene intermediate**; this activation enables elimination of water (P 16338409). Trapped catalytic-cycle snapshots of an inactive IGPD2 mutant show that substrate binding **switches an active-site Mn(II) between six- and five-coordinate states**, priming the site and facilitating the imidazolate intermediate (Bisson et al., 2015, P 26095028). This is a striking example of a metalloenzyme using controlled metal-coordination changes to drive discrete catalytic steps.
- **Evolutionary/bioinformatic support for the P. putida enzyme:** The 197-aa Q88R45 sequence contains the **two internally homologous, His-rich metal-binding motifs** diagnostic of IGPD — Region 1 `KGDT**H**IDD**HH**` (His62/His66/His67) and Region 2 `RG**H**NT**HH**Q` (His159/His162/His163). This intra-subunit two-fold pseudosymmetry reflects the **ancient gene duplication** that created the two metal-binding half-sites (P 16338409), so the *P. putida* enzyme is predicted with high confidence to use the identical dimanganese mechanism.

4. Biological Process and Pathway Context

Pathway: L-histidine biosynthesis from 5-phospho-α-D-ribose-1-diphosphate (PRPP) — KEGG map ppu00340 (Histidine metabolism) and module M00026 (PRPP ⇒ histidine). IGPD catalyzes **step 6 of 9** (UniProt/HAMAP MF_00076).

In *P. putida* KT2440 the pathway enzymes are encoded by a **distributed (non-operonic) set of genes**, mapped here via KEGG:

Step	Enzyme (EC)	KO	*P. putida* locus
1	ATP phosphoribosyltransferase (2.4.2.17), *hisG*	K00765	PP_0965
5	Imidazoleglycerol-phosphate synthase (4.3.2.10), *hisF*	K02500	PP_0293
6	**Imidazoleglycerol-phosphate dehydratase (4.2.1.19), *hisB***	**K01693**	**PP_0289**
7	Histidinol-phosphate aminotransferase (2.6.1.9), *hisC*	K00817	PP_0967
8	Histidinol-phosphate phosphatase (3.1.3.15)	K05602	PP_3157
9	Histidinol dehydrogenase (1.1.1.23), *hisD*	K00013	PP_0966

Monofunctional vs bifunctional HisB. KEGG module M00026 encodes two mutually exclusive branches for the dehydratase/phosphatase steps: (i) a **monofunctional IGPD (K01693)** plus a **separate histidinol-phosphatase**, or (ii) a **bifunctional IGPD/phosphatase (K01089)**, as found in enterobacteria. *E. coli hisB* is the bifunctional 355-aa enzyme (Chiariotti et al., 1986, P 3007936).

P. putida uses branch (i): the monofunctional 197-aa IGPD (PP_0289) works together with a **separate PHP-family histidinol-phosphate phosphatase (PP_3157, K05602)**. This definitively resolves that *hisB*/PP_0289 carries out **only** the dehydration step.

5. Subcellular Localization

The enzyme is **cytoplasmic** (UniProt Q88R45, HAMAP MF_00076). It has no signal peptide, no transmembrane segments, and no lipidation signal; the entire histidine biosynthetic pathway operates as a soluble cytosolic route. The catalytic reaction therefore occurs **in the bacterial cytoplasm**, where the enzyme functions as a large soluble homo-24-meric particle.

6. Biological / Translational Significance

- **Essentiality:** IGPD activity is required for histidine prototrophy; loss abolishes de novo histidine synthesis (histidine auxotrophy).
- **Drug/herbicide target:** Because the histidine pathway is absent in mammals, IGPD is an attractive, selective target. It is competitively inhibited by 3-amino-1,2,4-triazole (P 24311587) and is the target of triazole-phosphonate herbicides (P 16338409). Histidine metabolism/IGPD has also been implicated in bacterial biofilm formation (Zhou et al., 2018, P 29675012).

7. Supported and Refuted Hypotheses

Supported: - H1: *hisB*/PP_0289 is a monofunctional IGPD catalyzing step 6 of histidine biosynthesis (IGP → IAP + H_2O). — Supported by UniProt catalytic-activity annotation, KEGG K01693 mapping, and sequence length/domain content. - H2: The enzyme uses a conserved dimanganese active site and imidazolate/diazafulvene mechanism. — Supported by ortholog crystal structures (P 16338409 P 26095028 P 24311587) and conservation of the two His-rich motifs in Q88R45. - H3: The enzyme is cytoplasmic. — Supported by UniProt/HAMAP and absence of localization/membrane signals. - H4: In *P. putida* the phosphatase step is encoded separately (not by hisB). — Supported by KEGG module logic and identification of PP_3157 (K05602).

Refuted / ruled out: - *hisB* of *P. putida* is NOT the bifunctional IGPD+histidinol-phosphatase enzyme found in *E. coli* (197 aa single domain vs 355 aa two-domain; separate PP_3157 phosphatase present).

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8. Limitations and Future Directions

- **No direct P. putida experimental data:** Function is assigned by strong homology/annotation transfer and family-wide structural/mechanistic studies (Mtb, *Arabidopsis*), not by biochemical characterization of the *P. putida* enzyme itself. Enzyme kinetics (kcat/Km for IGP), metal dependence, and a *P. putida* IGPD structure remain to be determined experimentally.
- **Start-codon note:** UniProt annotates a 197-aa product; KEGG lists a 210-aa product with a 13-residue N-terminal extension. This is a start-site annotation discrepancy and does not affect the catalytic domain or function.
- **Regulation:** Transcriptional/feedback regulation of the *P. putida* his genes (e.g., attenuation, feedback inhibition of HisG by histidine) was not examined in detail and would refine the physiological picture.

References (PMIDs)

- Glynn SE, et al. Structure and mechanism of imidazoleglycerol-phosphate dehydratase. *Structure* 2005.

  P 16338409 .
- Bisson C, et al. Crystal structures reveal that the reaction mechanism of IGPD is controlled by switching Mn(II) coordination. *Structure* 2015.

  P 26095028 .
- Ahangar MS, et al. Structures of native, substrate-bound and inhibited forms of *M. tuberculosis* IGPD. *Acta Cryst D* 2013.

  P 24311587 .

- Chiariotti L, et al. Gene structure in the histidine operon of *E. coli*: identification and nucleotide sequence of the hisB gene. 1986.

P 3007936 .

- Zhou X, et al. Histidine metabolism and IGPD play a key role in cefquinome inhibiting biofilm formation. 2018.

P 29675012 .

- Database annotations: UniProt Q88R45 (HAMAP MF_00076); KEGG ppu:PP_0289 (K01693), module M00026; InterPro IPR000807/IPR020565/IPR038494; Pfam PF00475.
